

Welcome in!

Take a seat and **get comfortable**.
Make sure you have **VSCode downloaded** and a
GitHub account!

And, be sure to
connect with us on
social media.



Join the Kahoot for a
chance to win a
\$25 giftcard!



amazon



/SoDA

Intro to Spring Boot

“The tech-stack of Big Tech”

Goals:

- Understand how Spring Boot simplifies backend development.
- Build and run a REST API in under 30 minutes.
- Learn the connection between controllers, services, and repositories.



What is Spring Boot?

- A Java framework that makes it easy to create stand-alone, production-ready applications.
- Handles dependencies and configurations automatically.
- Built on top of the Spring Framework.
- Spring Boot is a backend framework



Why learn Spring Boot?

- Used by major companies (Amazon, PayPal, Amex).
- Reduces boilerplate setup compared to plain Spring.
- Perfect for building scalable, production-grade applications fast.



...

Compare Spring Boot to Just Frontend + Backend



Compare Spring Boot to Just Frontend + Backend

Frontend + Backend (Traditional):

- Backend handles routing manually
- You write SQL You manage serialization
- You configure everything yourself

Spring Boot (Modern):

- Automatic routing via annotations (@GetMapping, @PostMapping)
- ORM handles SQL for you
- Auto JSON serialization
- Dependency injection built-in Industry-ready microservices structure



Kahoot Time!



Questions 1-2



Spring Boot Basics: Structure

Spring Boot consists of:

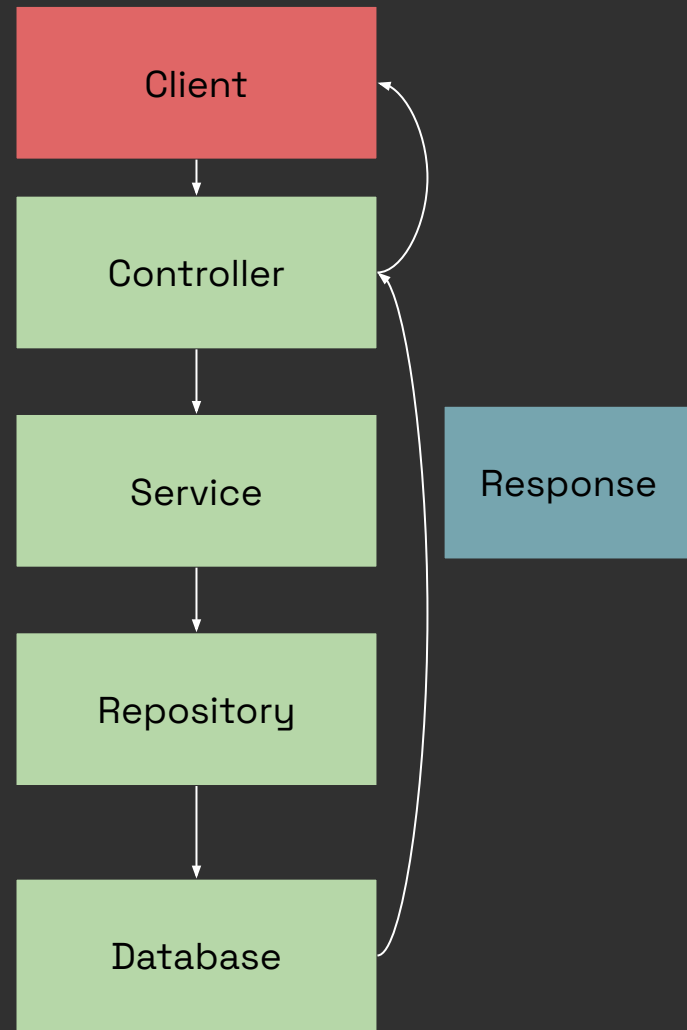
- Project Structure
 - src/main/java
 - src/main/resources
 - application.properties
 - Pom.xml

Why this matters: Spring Boot auto-configures 70–80% of boilerplate so you don't manually wire beans, servlets, or XML.

Spring Boot Basics: Core Components

- Core Components
 - Controller – handles API requests.
 - Service – holds business logic.
 - Repository – interacts with the database.
 - Model/Entity – represents data objects (tables).

Spring Boot Basics: Diagram



Spring Boot Basics: Hibernate and JPA

Why JPA?

- JPA is a standard that defines how to interact with databases.
- Hibernate is the most popular implementation of JPA.
- We use JPA so we don't write SQL manually, methods like `findAll()` are built-in.

Kahoot Time!



Questions 3-5



Controller Layer

Figure:

```
@RestController
@RequestMapping("/user")
public class UserController {

    @GetMapping
    public Collection<User> allUsers() {
        return users.values();
    }

}
```

Info

- @RestController → Combines @Controller + @ResponseBody.
- @GetMapping → Handles GET requests (read data).
- @PostMapping → Handles POST requests (create data).
- Example endpoints:
GET /users → Fetch all users
POST /users → Add a user

Service Layer

Figure:

```
@Service
public class UserService {
    @Autowired
    private UserRepository userRepository;

    public List<User> getAllUsers() {
        return userRepository.findAll();
    }

    public User addUser(User user) {
        return userRepository.save(user);
    }
}
```

Info

- @Service → Marks the class as part of the business logic layer.
- Contains reusable methods (e.g., retrieving or saving users).
- Keeps controllers clean and focused on routing.

Repository Layer

Figure:

```
public interface UserRepository extends  
JpaRepository<User, Long> {  
    List<User> findByFavoriteColor(String  
favoriteColor);  
    List<User> findAll (String findAll)  
}
```

Info

- JpaRepository gives CRUD operations like `findAll()`, `save()`, `deleteById()` automatically.
- You can define custom finder methods (e.g., `findByFavoriteColor`) just by naming convention

DAO Class

Figure:

```
import jakarta.persistence.*;
import lombok.Getter;
import lombok.Setter;

@Entity
@Getter
@Setter
public class User {

    @Id
    @GeneratedValue(strategy = GenerationType.IDENTITY)
    private Long id;

    private String username;
    private String favoriteColor;
}
```

Info

- @Entity → Maps this class to a table in the database.
- @Id → Primary key column.
- @GeneratedValue → Auto-increments the ID.

What is a Java Bean?

Real-World Relevance

- Spring Boot is used in microservices architectures at Amazon, Netflix, and Google.
- Common in technical interviews for backend roles.
- Demonstrates understanding of REST APIs, dependency injection, and scalable architecture.

Skills gained: REST design, ORM, and backend integration.

Kahoot Time!



Questions 5-13



Hands-On Demo



We'll show:

- The Structure of Springboot
- How it all works together in real time
- How responses are made to the client

Starting a Spring Boot Project

spend hours
setting up Java

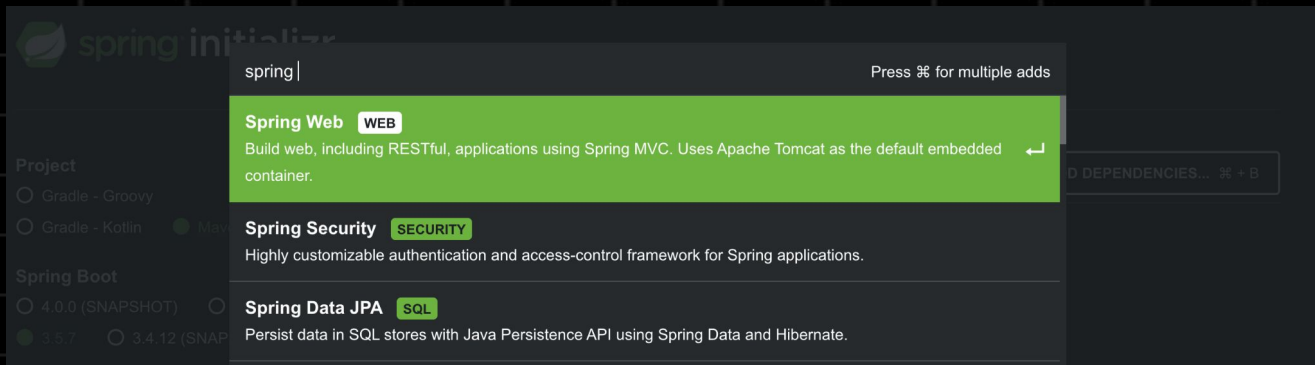
git clone



Standard Way of Starting Spring



- start.spring.io (do not use, YET)
 - Select 'Maven' and 'Java'
 - Click on 'Add Dependencies' and add 'Spring Web'
 - Click on 'Generate', download and unzip

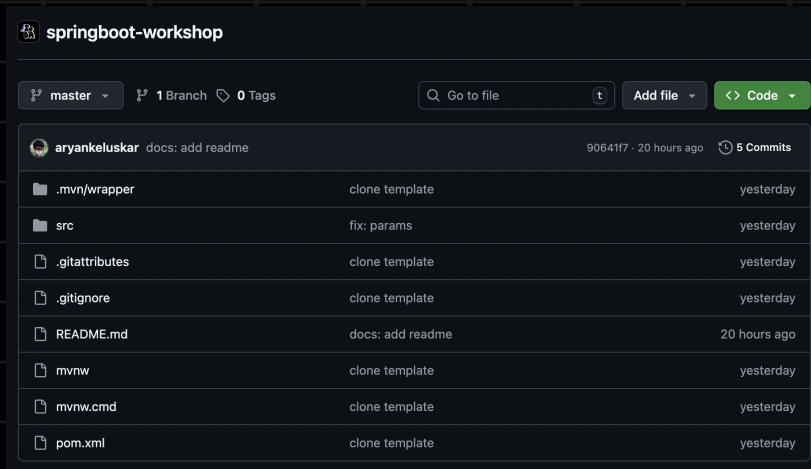


But, git clone for faster setup

- git clone

<https://github.com/asusoda/springboot-workshop.git>

Posted in Discord! See #announcements



Let's understand APIs first



What is an API?

What is an API?



What is an API?

The screenshot shows the 'My ASU' student portal interface. At the top, the ASU logo and 'My ASU' title are visible. Navigation links include 'Finances', 'Campus Services', 'Profile', and 'Help'. A row of service icons includes PeopleSoft, Gmail (6818), Canvas, Google Drive, ASU Library, and My Apps. The 'My Classes' section displays a semester selector with 'Spring '26' highlighted, a list of courses (APM 527, CSE 450, CSE 486), and a 'View My Schedule' button. Below this is the 'My Agenda' section, showing the date 'Monday, November 17, 2025' and a list of tasks including 'BIO 493 Honors Thesis', 'CSE 294 Special Topics', 'CSE 485 Computer Sci Capstone Proj I', 'CSE 475 Found of Machine Learning' (with time and location), and 'CSE 578 Data Visualization'. A user profile dropdown menu is open, showing the user's name 'Aryan Keluskar', a profile picture of a blue cat, and details: Primary Name: Aryan Keluskar, Preferred Name: Aryan Keluskar, ASU ID (emplid): 6767676767, and ASURITE User ID: akeluska. The dropdown also includes a 'View Profile' button and links for 'New Features', 'Help', and 'Feedback'. The main content area on the right includes sections for 'My Programs', 'Barrett Honors College Honors Progress', 'Academic Support Team', and 'Academic Advising'.

ASU Home My ASU Colleges and Schools Aryan (Sign Out)

ASU My ASU
Arizona State University

Finances Campus Services Profile Help

PeopleSoft Gmail 6818 Canvas Google Drive ASU Library My Apps Calendar

My Classes

Fall '24 Spring '25 Summer '25 Fall '25 **Spring '26** All Semesters ▾

APM 527 Deep Neural Networks
CSE 450 Design&Analysis of Algorithms
CSE 486 Computer Sci Capstone Proj II

Motsch
Daymude
Chavez Echeagaray, Parupudi

View My Schedule

Registration > Class Search Books Grades & Transcripts

My Agenda

Monday, November 17, 2025 < Today >

BIO 493 Honors Thesis

CSE 294 Special Topics

CSE 485 Computer Sci Capstone Proj I

CSE 475 Found of Machine Learning
9:00 AM - 10:15 AM | Tempe - PSH152

CSE 578 Data Visualization

Full agenda Download calendar

Aryan Keluskar

Primary Name: Aryan Keluskar
Preferred Name: Aryan Keluskar
ASU ID (emplid): 6767676767
ASURITE User ID: akeluska

View Profile

New Features
Help
Feedback

My Programs

Barrett Honors College
Honors Progress

Add minor Add certificate

Academic Support Team

Academic Advising
Degree requirements, graduation progress, and course registration.

Tutoring
Find assistance with writing or a specific subject area

Ask a Librarian
Get library help over chat, email, phone, FAQs or in person.

What is an API?

The screenshot displays the MyASU (Arizona State University) web portal. At the top, the ASU logo and 'My ASU' title are visible. Navigation links include Finances, Campus Services, Profile, and Help. A user profile overlay for 'Aryan Keluskar' is shown, featuring a blue cat avatar and details such as Primary Name, Preferred Name, ASU ID, and ASURITE User ID. A 'View Profile' button is also present. A large text overlay in the center reads 'How does MyASU know what to show?'. The background shows sections for 'My Classes' (listing courses like APM 527, CSE 450, and CSE 486) and 'My Agenda' (showing dates and class times).

ASU Home My ASU Colleges and Schools Aryan (Sign Out)

ASU My ASU
Arizona State University

Finances Campus Services Profile Help

PeopleSoft Gmail 6818 Canvas Google Drive ASU Library My Apps Calendar

My Classes

Fall '24 Spring '25 Summer '25 Fall '25 **Spring '26** All Semesters ▾

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Motsch
Daymude
Chavez Echeagaray, Parupudi

[View My Schedule](#)

Registration > Class Search Books Grades & Transcripts

My Agenda

Monday, November 17, 2025 < Today >

BIO 49
CSE 25
CSE 485 Computer Sci Capstone Proj
CSE 475 Found of Machine Learning
9:00 AM - 10:15 AM | Tempe - PSB
CSE 578 Data Visualization

Full agenda Download calendar

Aryan Keluskar

Primary Name: Aryan Keluskar
Preferred Name: Aryan Keluskar
ASU ID (emplid): 6767676767
ASURITE User ID: akeluska

[View Profile](#)

New Features
Help
Feedback

My Programs

Programs

Computer Degree Program

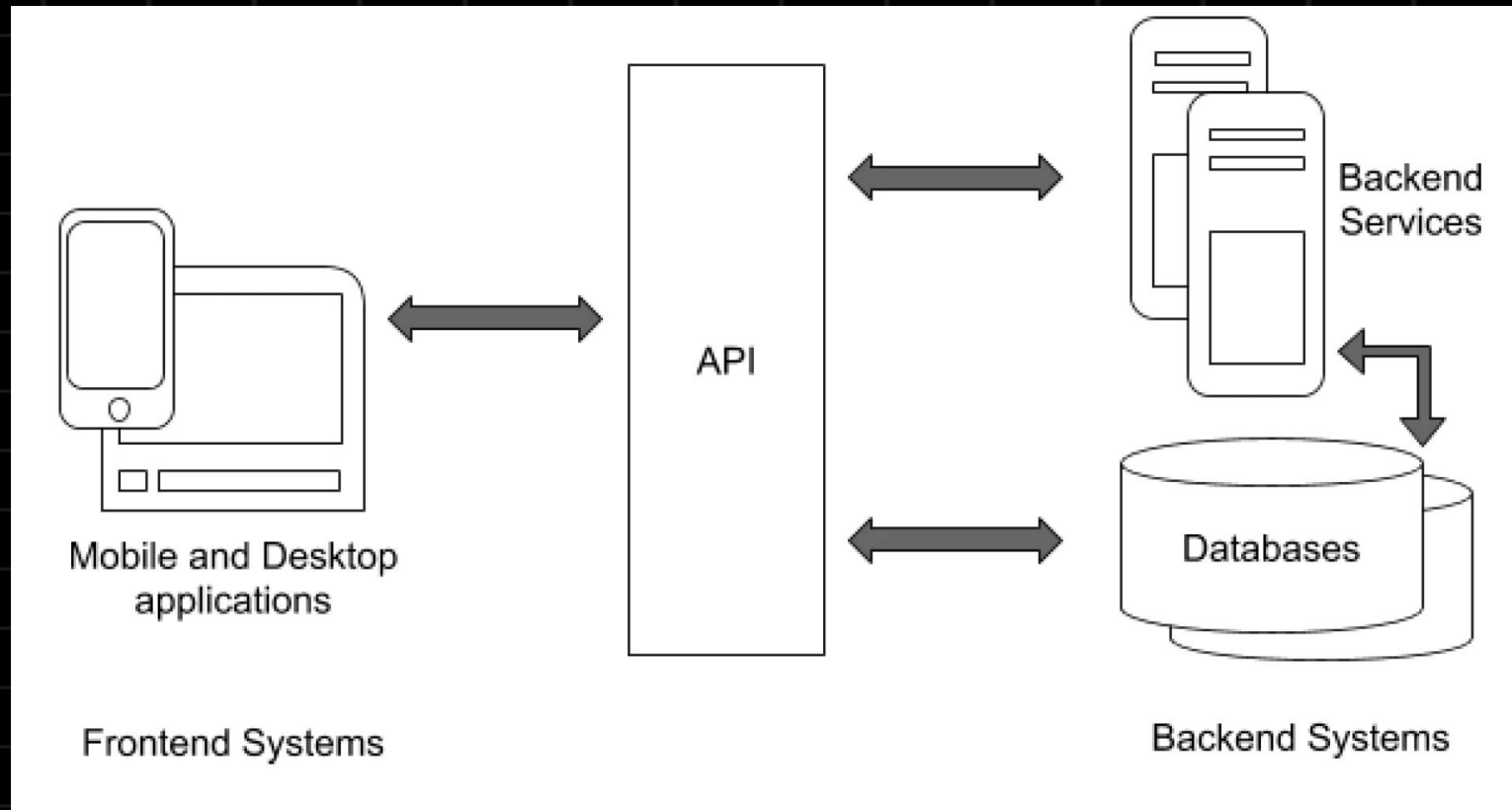
and course registration.

tutoring
Find assistance with writing or a specific subject area

Ask a Librarian
Get library help over chat, email, phone, FAQs or in person.

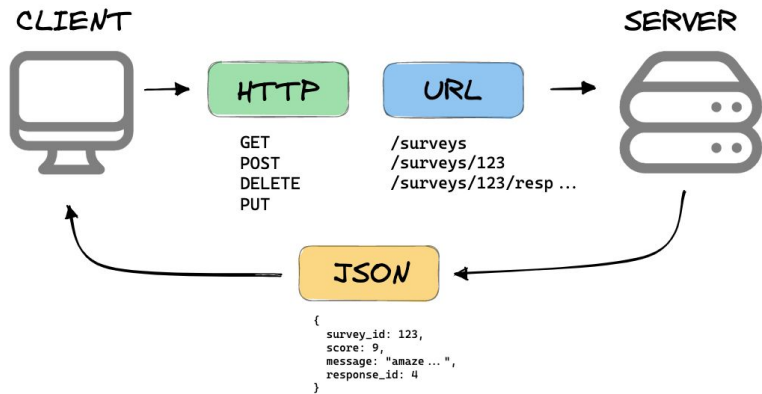
How does MyASU know what to show?

What is an API?



RESTful API Design

WHAT IS A REST API?



mannhowie.com

Flashback to Databases

CRUD Operations

Create | Read | Update | Delete

```
-- Create
INSERT INTO Students (id, name, major, gpa)
VALUES (1, 'Alex', 'Computer Science', 3.9);

-- Read
SELECT * FROM Students WHERE gpa > 3.5;

-- Update
UPDATE Students SET gpa = 4.0 WHERE id = 1;

-- Delete
DELETE FROM Students WHERE id = 1;
```



Relation to API Design

CRUD Operations → **REST**ful API

Create	POST
Read	GET
Update	PUT
Delete	DELETE

Relation to API Design

CRUD Operations → **REST**ful API

Create	POST
Read	GET
Update	PUT
Delete	DELETE

Unlike SQL that
returns Tables,
an API is
expected to
return JSON!

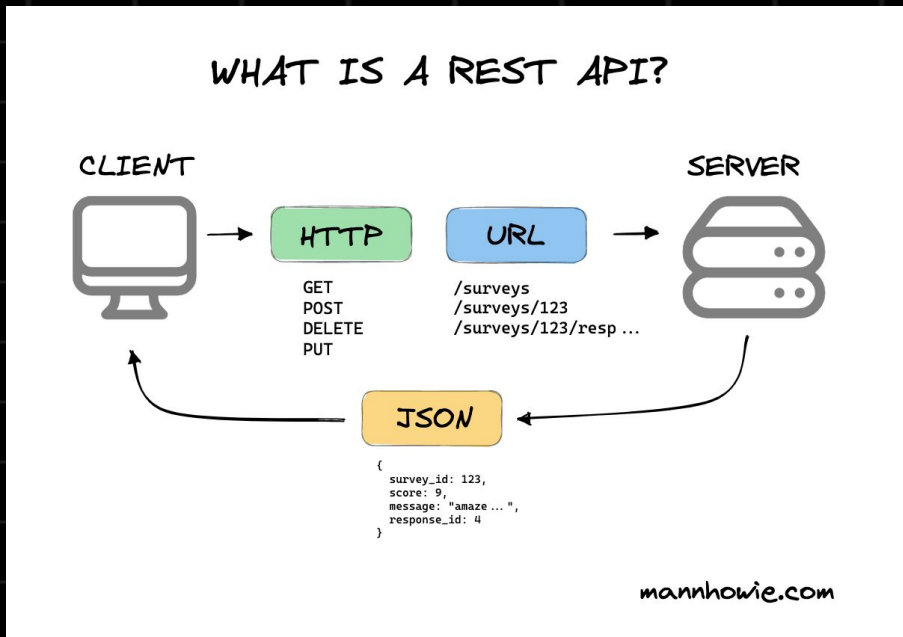
Relation to API Design

CRUD Operations → **REST**ful API

Create	POST
Read	GET
Update	PUT
Delete	DELETE



What are we building?



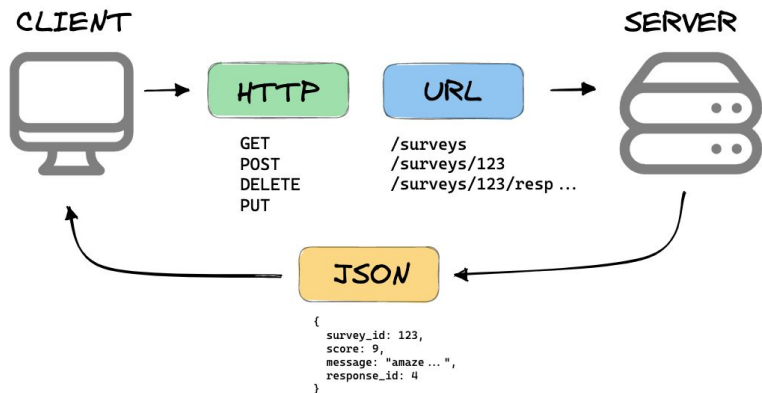
GET /user

```
[
  {
    "name": "Aryan Keluskar",
    "asurite": "akeluska",
    "number": "6767676767"
  },
  {
    "name": "Sun Devil",
    "asurite": "new_sparky",
    "number": "3689071223"
  },
  {
    "name": "Sun Devil",
    "asurite": "sparky",
    "number": "3689071223"
  }
]
```

What are we building?

GET /user

WHAT IS A REST API?



mannhowie.com

My ASU Colleges and Schools **Aryan** (Sign Out)

Aryan Keluskar ^

Aryan Keluskar

Primary Name: Aryan Keluskar

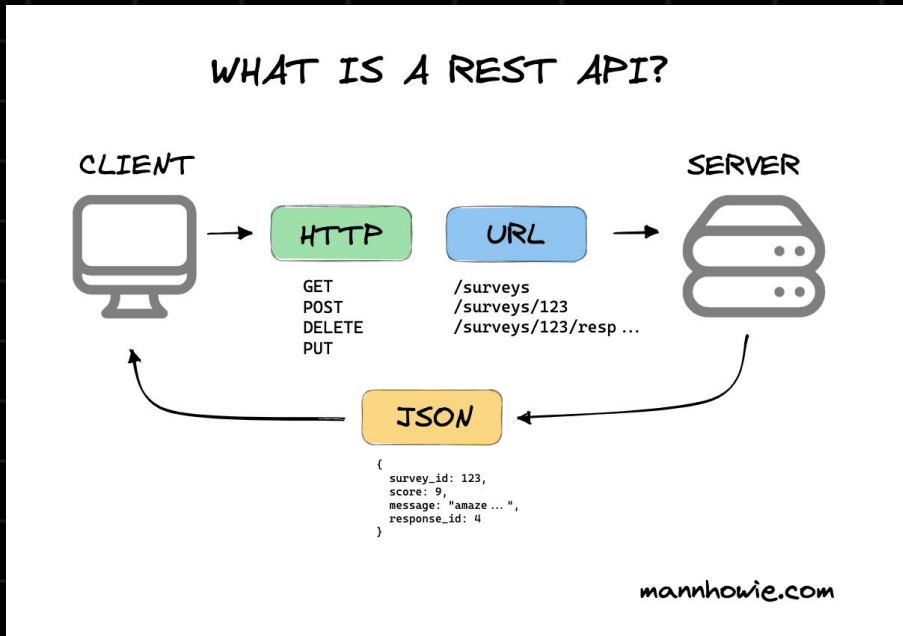
Preferred Name: Aryan Keluskar

ASU ID (emplid): 6767676767

ASURITE User ID: akeluska

[View Profile](#)

What are we building?



GET: does not allow JSON body
POST/PUT/DELETE: allow (and even encourage) JSON

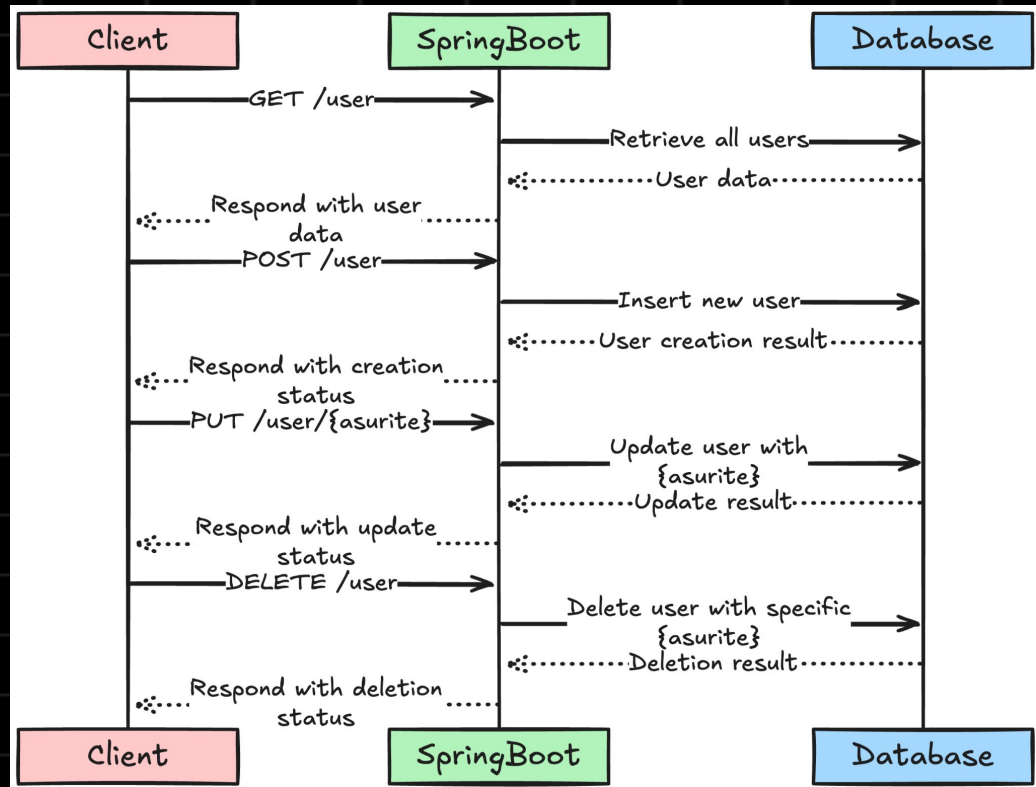
Standard Browsers only show you GET requests. For POST and other, you use Postman or Curl

The Routes in today's demo

READ all users (GET /user)

CREATE a user (POST /user)

```
{  
  "asurite": "sparky", "name": "Sun Devil",  
  "number": "3689071223"  
}
```



The Routes in today's demo

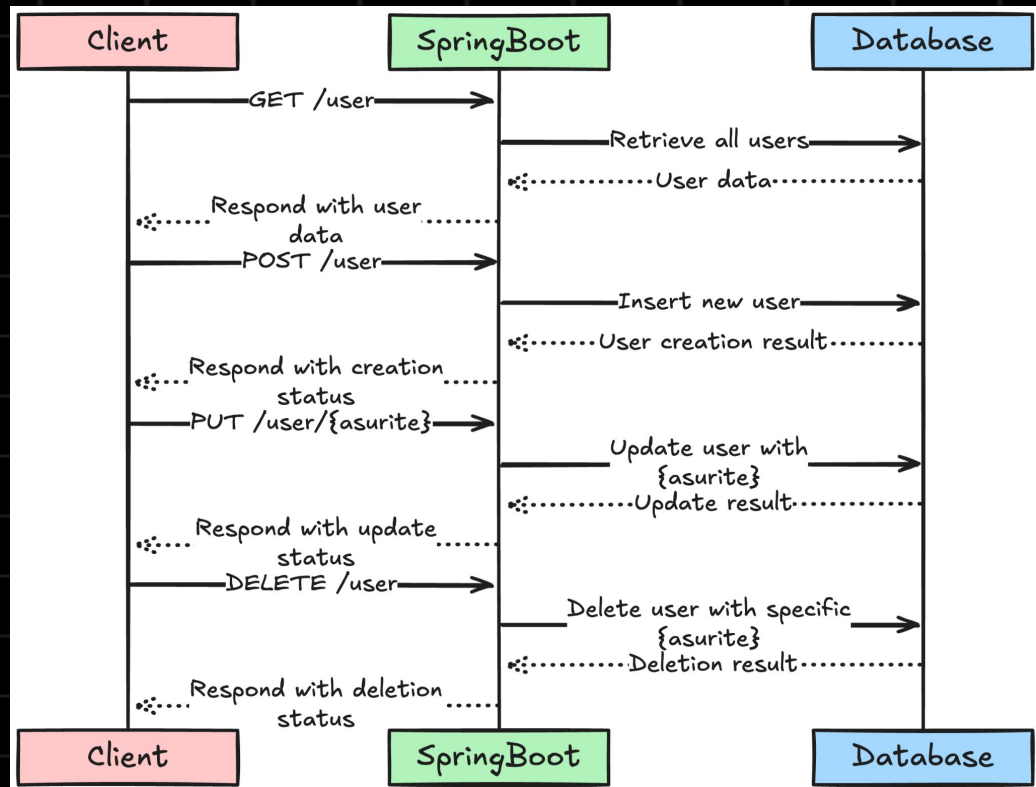
UPDATE a user (PUT /user/{asurite})

```
{  
  "name": "Sparky the Sun-Devil",  
  "number": "3689071223"  
}
```

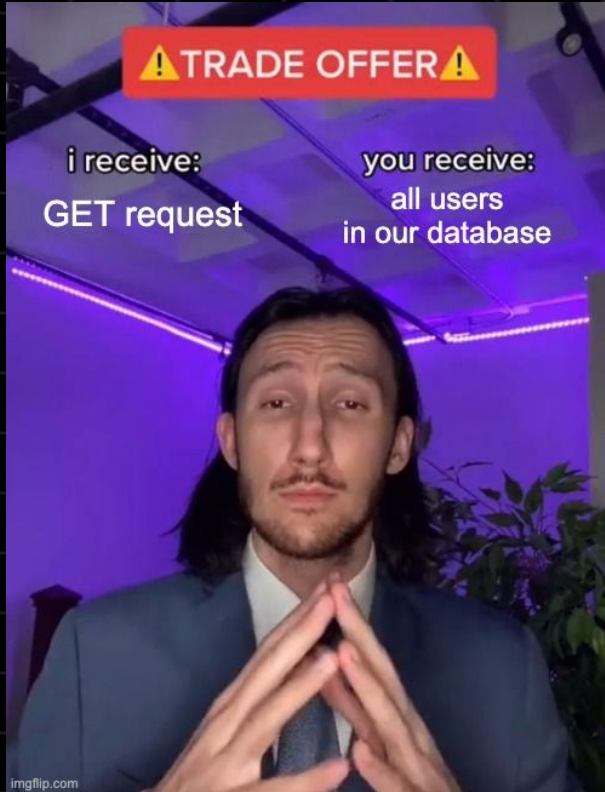
DELETE a user (DELETE /user)

```
{  
  "asurite": "sparky"  
}
```

These are already setup!



Basically,

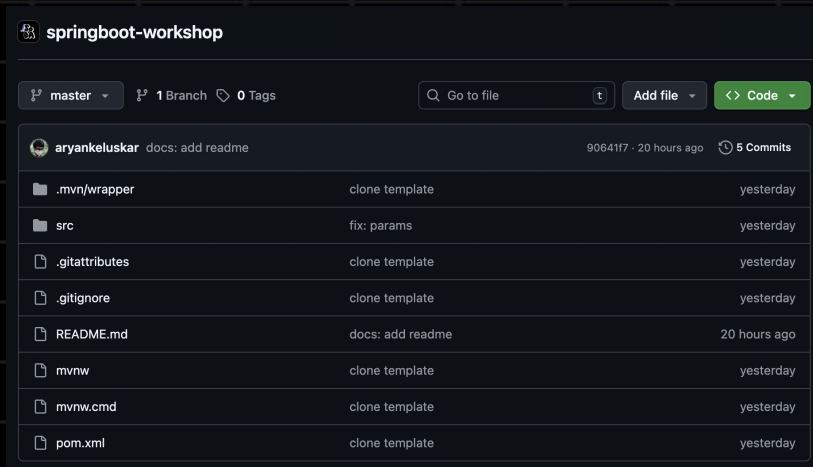


Let's Begin!

- git clone

<https://github.com/asusoda/springboot-workshop.git>

Posted in Discord! See #announcements



That's all!

Any questions?

